

Lin Ju

Boston, MA (Open to Relocate) | 617.756.9636 | ju.li@northeastern.edu | LinkedIn | GitHub | Portfolio

Education

Northeastern University

Master of Science in Computer Software Engineering

Boston, MA

Expected Dec 2026

Tongji University

Bachelor of Engineering in Mechanical Engineering (Second Major: German)

Shanghai, China

Sept 2016 - Jul 2021

Work Experience

Founding AI Engineer Intern

Jun 2026 - Aug 2026

Interlock Systems Inc. (Harvard startup)

Boston, MA

- **Founding engineer** on a multi-agent AI platform that automates manufacturing workflows from engineering documents into ERP/MRP systems; delivered the \$24K Stage 1 customer deployment and onsite rollout.
- **Built a reference SAP S/4HANA ERP** with SAP CAP (Node.js): SAP-identical OData APIs and a 12-app Fiori launchpad with live KPIs, plus Python ETL pipelines loading customer data into PostgreSQL, deployed on AWS EC2 as a hardened systemd service.
- **Shipped an on-prem React and FastAPI app** on a Fortune 500 chemical manufacturer's production ERP, replacing hours of manual screen work for project managers who now reschedule deliveries and revise work orders daily; natural-language search routes queries to deterministic SQL at 97.5% accuracy, with every write reversible via logged undo.
- **Built a CP-SAT (Google OR-Tools) scope and schedule optimizer** for plant turnaround planning, with an LLM copilot that converts plain English into typed constraints, re-solves, and reports before/after metrics for approval.
- **Owned platform architecture and delivery:** document ingestion and OCR/parsing pipelines, LLM evaluation and automated test frameworks, Docker, and CI/CD.

Software Developer Intern (Spark RAPIDS ML, Holoscan & Isaac)

Jun 2025 - Dec 2025

Nvidia

Shanghai, China

- **Designed and built backend data storage utilities and API migration tools** for GPU-accelerated workloads in Python, Docker, and Linux, streamlining internal developer workflows.
- **Benchmarked GPU-accelerated deep learning pipelines against CPU baselines**, profiling performance and validating accuracy to guide optimization decisions.
- **Built a test automation system to improve code coverage**, including a Java-based JaCoCo execution-data reader, a Python MCP server using FastMCP/OpenAPI, and IDE integration, raising code coverage to 80% through AI-assisted test generation.
- **Integrated automation into Jenkins CI/CD pipelines**, debugging failures across the Holoscan and Isaac C++ SDKs and catching performance regressions before release.

Data Engineer

Sept 2023 - Aug 2024

Payoneer Inc.

Shanghai, China

- **Built data integration and automation workflows** using Google Cloud Platform, BigQuery, and Python for a new product initiative, contributing to a 20% increase in ROI.
- **Developed an e-contract platform** using Salesforce and HTML/CSS/JS for a better contract signing procedure under compliance regulations.
- **Developed an all-in-one internal app** for marketing campaign tracking and business development actions.

Data Engineer

Dec 2021 - Sept 2023

RDM Management Consulting

Shanghai, China

- **Identified business pain points and built data dashboards** using SQL Server and Python to support decision-making and operational improvements.
- **Designed and implemented data modeling solutions**, including a fully automated data crawling pipeline in Python and SQL, and built machine learning models for cluster analysis and consumer segmentation to predict consumer behavior, producing a cost-efficient system and a 50% year-over-year increase in ROI.
- **Implement a concise data warehouse** and designed the ETL workflow in the database.

Academic Projects

RaGuard: Vulnerability Detection and Mitigation Copilot Using RAG-Augmented Local LLMs (*under review*)

2026

- **Built an agentic AI security scanner in Python** where an LLM orchestrator routes specialist small language models through a tool-calling loop with JSON-schema structured outputs, grounded by retrieval-augmented generation (RAG) over a 994-chunk CWE/CVE knowledge base; findings are confirmed only after ffuf reproduces them against a live containerized target.
- **Fine-tuned a 1.7B specialist model with QLoRA (Unsloth, PyTorch)** on an 820-example corpus distilled from 11.8K logged teacher-LLM calls, then merged and quantized it to GGUF for local serving; retained 80% of the 8B baseline F1 at 2x throughput and \$0 inference cost.
- **Built an LLM evaluation harness** benchmarking three model arms on precision, recall, F1, latency, and token cost, with config-driven model routing in one versioned YAML; ran the full stack locally on Ollama in two OWASP-hardened Docker containers.

Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks [Paper]

2026

- Co-authored a study of how multimodal large language models perform visual reasoning on referring expression tasks.
- Accepted at the International Conference on Learning Representations (ICLR 2026).

Skills

Languages: Python, Java, C++, SQL (BigQuery, T-SQL, MySQL), JavaScript/TypeScript, Bash/Shell

AI & LLM Systems: RAG, Retrieval & Embeddings, LLM Fine-tuning (LoRA/QLoRA, PEFT), Multi-Agent Systems, Tool Calling, MCP, Prompt Engineering, LLM Evaluation, DSPy, Ollama, Groq, Hugging Face Transformers, TensorFlow, Scikit-learn

Tools & Cloud: Docker, AWS (EC2, S3, Secrets Manager), Kubernetes, Git, GitHub Actions, Jenkins CI/CD, Linux, Cloudflare Tunnel, Pytest, Behave

Backend & Data: FastAPI, React, Node.js, NestJS, Spring Boot, REST APIs, PostgreSQL, SQL Server, MongoDB, SQLite, Pandas, NumPy, ETL Pipelines